

Electrical Topology Diagram (Implementation v6)

As-of date: 2026-08-12

Purpose: provide a complete, install-level electrical topology for the current build scope, including all major electrical components, fuse IDs, fuse housings, planned wire gauges, and estimated one-way run lengths for procurement planning.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Detailed fuse matrix: docs/implementation/ELECTRICAL_fuse_schedule.md
- Battery and trunk recalculation record: docs/studies/ELECTRICAL_battery_fuse_wire_recalc_2026-02-18.md
- Decisions and unresolved items: docs/core/TRACKING.md
- Procurement source of truth: bom/bom_estimated_items.csv

Sweep Outcomes Included In This Revision

- Keeps alternator charging architecture on the dedicated 48V secondary alternator path (Mechman + WS500 + APM-48 baseline); obsolete pre-Mechman charger paths are removed from primary topology.
- Keeps Lynx Slot 3 branch to alternator input with F-04 200A/80V MEGA.
- Removes obsolete pre-Mechman engine-bay fuse/conductor placeholders from active architecture.
- Clarifies the confirmed PH-VAN harness as one 15A high-interrupt in-line-fused combined power/positive-sense red lead (F-12/F-13-PHVAN) and locks the unfused purple/grey pair to the Wakespeed shunt on the battery side of the hard-mounted SmartShunt in the common battery-negative path.
- Adds Ford Upfitter #3 -> F-15 -> WS500 brown ignition manual alternator-control path.
- Adds the detailed Mechman/WS500/APM-48 install guide as the shop reference for staged installation, first-run checks, and load-dump/shutdown handling.
- Defines APM-48 as a parallel surge clamp at the alternator rather than a series charge-current device.
- Added explicit fuse-holder/housing definitions for every fuse family (Class T , Lynx MEGA , inline MIDI/ANL/AMI , PV gPV , and AT0/ATC).
- Added conductor schedule across 48V , 12V , PV, and AC segments with explicit assumptions.
- Updated 12V topology to a shared 12V junction fed by an Orion-Tr Smart 48/12-30 charger and a 12V 100Ah buffer battery branch, with F-11 source fuse plus SW-12V-BATT manual isolation.
- Added a full-circuit estimated run-length validation pass (C-01 through C-40) and purchase-ready wire rollup totals.

Current Commissioning Snapshot (2026-08-02)

- 48V bus live-tested at 55.5V throughout the system, including at the MultiPlus.
- MultiPlus-II inverter mode tested with inverter light on, slight normal hum, and no reported errors.
- SmartShunt and Orion-Tr Smart connected in VictronConnect.
- Cerbo GX access point/remote-console workflow active; Cerbo is powered from a small inline fused 48V feed and connected to MultiPlus via VE.Bus RJ45.
- Short AC-in shore-charge test passed at household-source current limits: about 1294W shore input and about 54.3V x 21.6A battery charging in bulk.
- MultiPlus LiFePO4 charge profile is programmed/owner-verified by supervised first-battery behavior. Owner reports the Orion/ 12V buffer path now charges and operates correctly. Hold open: permanent-path shore dead checks/first-use acceptance, full-bank charge/rest/match/parallel closeout, AC-out branch/GFCI, alternator commissioning, Cerbo hard-mounting, and final strain-relief/abrasion-control.

Current Physical Installation Snapshot (2026-08-27)

- The electrical module is hard-mounted through the finished Lonseal/plywood floor into registered truck-bed hardpoints and tied into the hard-mounted Bench/Galley extrusion structure. Owner reports the integrated assembly is extremely stiff; remaining mobile restraint includes batteries, cooler, air fryer, bench lid, monitor/laptop hardware, rigid cargo, and terminal/cable protection.
- The driver-rear shore inlet and exterior-to-module cable route are installed. One accessible, enclosed, conductor/gauge-rated three-wire L/N/PE splice into the AC-input side remains before dead checks and controlled energization.
- All three batteries' positive and negative 2/0 AWG branch cables are cut, lugged, heat-shrunk, and landed at the battery-side busbars. Battery 1 completed the corrected isolated charge cycle, Battery 2 resumed and reached absorption on 2026-08-03 , and Battery 3 remains pending; keep all three isolated until their rested voltages are matched within 0.1V before paralleling.
- The 12V buffer-battery/Orion branch is owner-reported operational. Keep its negative direct to the fuse-panel main negative stud, not through SW-12V-BATT ; the positive path remains F-11 -> SW-12V-BATT -> panel main + .

- Owner report 2026-08-27 : Starlink Standard 4 X is wired, switched, and working from the direct-DC branch. The ICECO/fridge and water pump are wired and switched; loaded functional proof remains part of branch closeout. Starlink uses a dedicated 20A blade fuse. Three separately fused charging outlets are installed: driver-side 100W and two passenger-side 65W . Preserve exact slot/fuse labels, polarity, terminal covers, loaded voltage-drop/heat tests, and final switch/service-face mounting.
- Owner report 2026-08-27 : C-31 Desk/office and C-32 Galley/passenger branches each feed one origin GFCI plus one downstream duplex, for four duplex devices/eight plug positions total. Before routine use or closure, prove breaker values, 12 AWG , branch separation, AC LINE/LOAD/PE , box/clamp/fill, downstream protection from each GFCI LOAD , trip/reset in inverter and accepted-shore modes, and representative load behavior.
- The current narrow switch bundle is temporary and not acceptable as a finish state. De-energize correctly, individually insulate/label DC switch terminals, and mount them on a removable plywood/ABS service face. Any adjacent 120V receptacle stays in its own listed covered box with cable clamps and PE continuity; do not mount exposed AC device backs in the DC switch cavity.

Orion fuse discriminator — one input fuse only

- Lynx Slot 4 / F-05 : **Orion 48V input positive**, one verified 40A MEGA (58VDC minimum under the locked 56.8V charge ceiling; Victron CIP138040020 40A/80V is the replacement fallback) feeding the existing 6 AWG directly. This is the final single input fuse.
- F-06 : **retired standalone Orion input-fuse position**. Do not install MIDI, FKS/ATO, DIN, or a second fuse after Slot 4.
- Lynx Slot 2 / F-03 : **MPPT branch**, 60A/80V MEGA. If the physically X-marked/misrated Lynx fuse is in Slot 2, replace it with the purchased Victron 60A/80V MEGA; it is not the Orion input fuse.
- F-07 : **Orion 12V output positive**, 60A/80V MEGA in the separate Victron holder near the Orion.
- F-11 : **12V buffer-battery positive**, 100A ANL near the battery, upstream of SW-12V-BATT .
- No 32V -rated fuse is acceptable on any energized 48V branch. Identify the physical X-marked fuse by Lynx slot before replacing or discarding it.

Length Estimation Defaults Used In This Pass

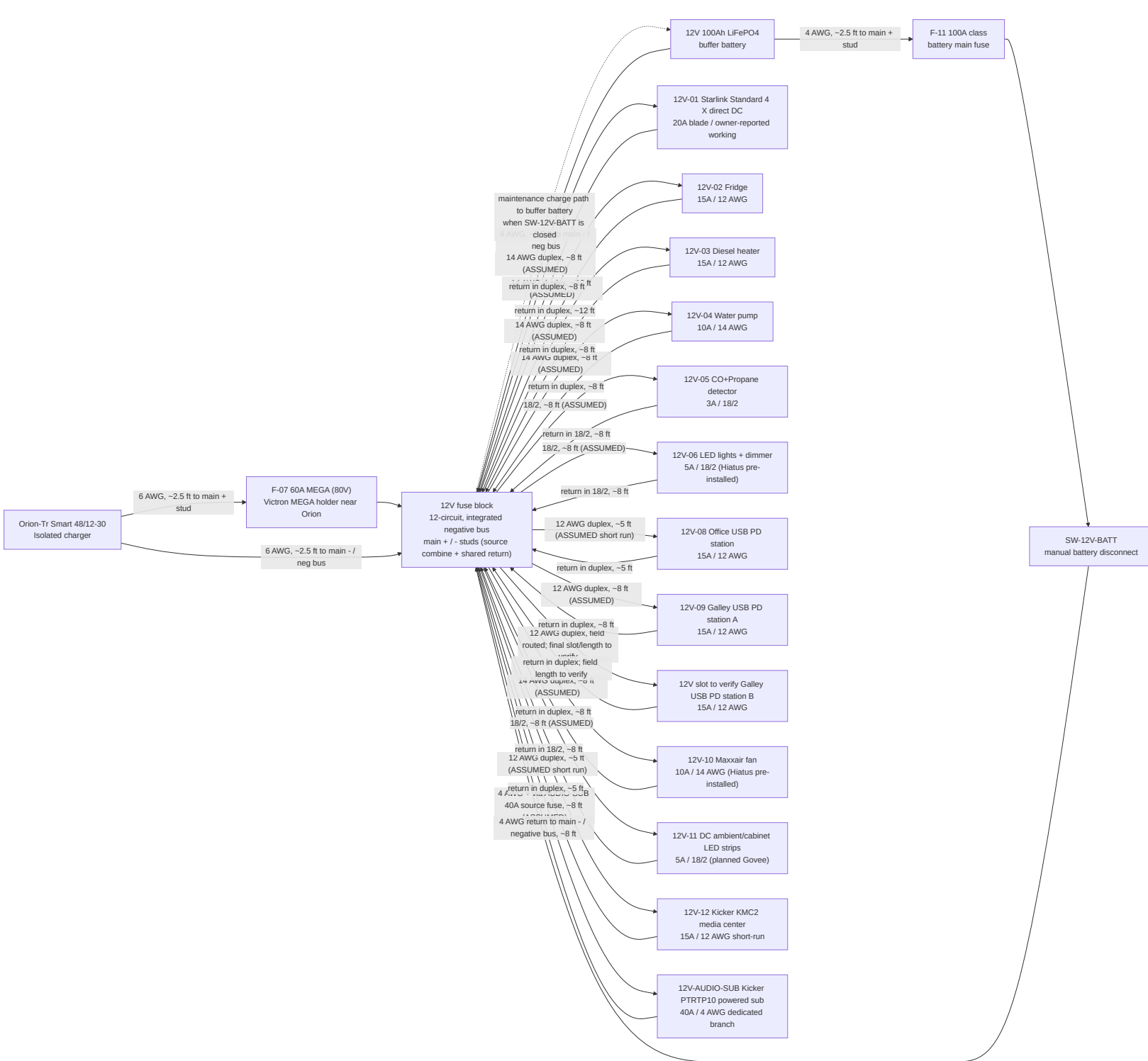
1. Cabinet internal interconnect default: 2.5 ft one-way (ASSUMED).
2. Cabinet-to-near load branch default: 8 ft one-way (ASSUMED).
3. Cabinet-to-far load branch default: 12 ft one-way (ASSUMED).
4. AC branch to receptacle chain default: 15 ft one-way per branch leg (ASSUMED).
5. Policy lock: use the smallest gauge that meets current and voltage-drop targets; do not auto-upsize, but flag warnings when margin is tight.
6. Parallel battery bank lock: keep each battery's **total positive + negative path resistance** similar. Equal positive-only length is not required if the positive-length differences are offset by negative-length differences; do not add unnecessary 2/0 cable coils solely for cosmetic equality.

Battery Fuse/Wire Recalculation Basis (2026-02-18 + 2026-03-19 sync)

- Scope in this pass is limited to battery-side and major 48V trunk paths (C-01 through C-15).
- Provisional battery listing inputs used: 51.2V 100Ah , <=200A current limit per battery.
- Conservative sizing factors used in this pass:
 1. Parallel-sharing factor $K_{share} = 1.5$
 2. Continuous margin factor $K_{cont} = 1.25$
- Current envelope used for battery-discharge branch sizing: $I_{total} = F-02 + F-05 = 125A + 40A = 165A$.
- Per-battery design current: $I_{batt_design} = (165A / 3) * 1.5 = 82.5A$.
- Continuous-adjusted minimum battery branch fuse threshold: $82.5A * 1.25 = 103.1A$.
- Provisional battery branch fuse selection: F-01A/B/C = 200A Class T , constrained by the provisional battery <=200A current-limit listing.
- Final lock gate: validate true 51.2V battery datasheet/manual current and terminal limits before permanent fuse lock; if lower limits are confirmed, move to 175A .
- Cable procurement remains estimate-based until CAD/field run lengths are frozen. This pass sets a no-padding 2/0 estimate baseline of 77.5 ft total (42.5 ft red, 35.0 ft black), including the dedicated alternator migration path.

Complete Power Topology (48V Core + Charge Sources)



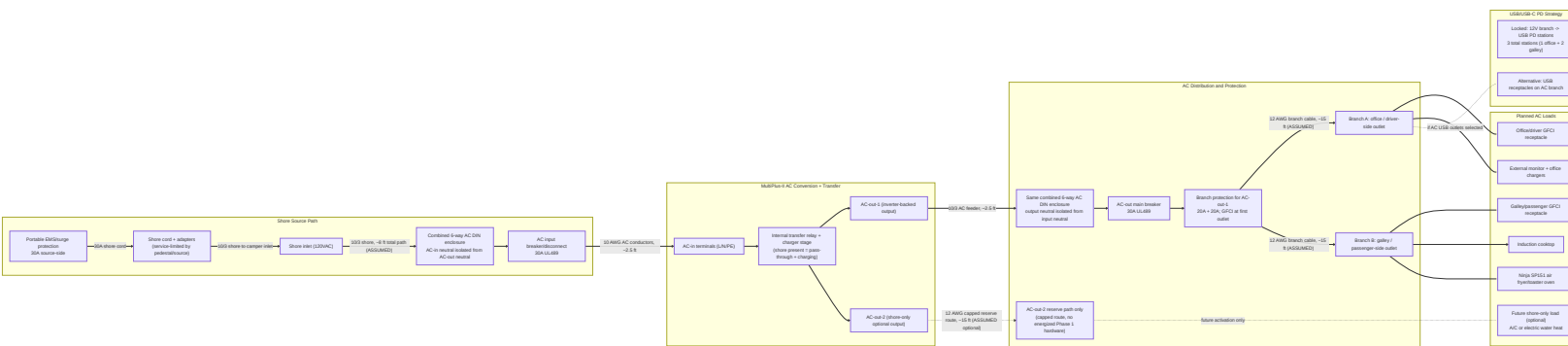


12V Operating Intent (Locked)

- Orion-Tr Smart 48/12-30 is the primary 12V charger/feed source.
- SW-12V-BATT is **normally closed** in operation; open is service/isolation mode only.
- With SW-12V-BATT closed, Orion output at the fuse-block main + stud maintains/charges the 12V buffer battery through the shared-junction path.
- With SW-12V-BATT open, the buffer battery is isolated from the main + stud, but the fuse-panel bus can remain powered from the Orion while the Orion is enabled. For service isolation, turn active loads off, disable the Orion and confirm off/not charging, then open SW-12V-BATT and meter the bus before work.
- Replace the Orion's always-on L-H jumper with a maintained SPST dry-contact remote switch. Startup closes SW-12V-BATT before enabling the Orion; normal shutdown disables the Orion before opening the battery switch.
- The buffer battery remains in the active operating path during normal use and is intended to absorb transients/peaks on the 12V rail.
- The fuse block is the 12V junction device in this baseline: main + stud is the source-combine point, and the integrated negative bus/main - is the shared return point.
- Do not solder-splice high-current source conductors; terminate with crimped lugs on rated studs/junction hardware.

- Hiatus pre-installed 12V branches are tracked here as 12V-06 (factory LED lights with dimmer) and 12V-10 (Maxxair fan); verify final installed branch labeling during electrical audit.
- Planned Govee/ambient strip lighting is a separate branch (12V-11) and not part of the Hiatus factory 12V-06 lighting circuit.
- DC-first lighting intent: keep ambient/cabinet strip lighting on 12V-11 so nighttime lighting does not require inverter operation.
- Camper audio intent: Kicker 46KMC2 source unit uses a 15A 12V-12 branch from the fuse block, while Kicker 49PTRTP10 powered sub uses a separate 40A fused 4 AWG branch from the 12V source/junction rather than a normal blade-fuse panel branch. Bass peaks may exceed the Orion 30A continuous feed and are supported by the 12V buffer battery for short durations.

AC Path Topology (Shore + Inverter Output, Full Hierarchy)



AC Operating Behavior (Expected)

- Shore present: MultiPlus transfer relay closes, AC-in is passed to AC-out paths, and charger stage charges the 48V bank.
- Shore absent: MultiPlus transfers to inverter mode and powers AC-out-1 from battery; AC-out-2 drops by design.
- AC-in hardware is 30A (source/adapters -> portable EMS -> shore cord -> L5-30 inlet -> 30A breaker -> 10 AWG AC-in conductors); set the MultiPlus limit to source-specific permitted draw— 10A for first household tests and 12A maximum policy on a normal 15A outlet, or the accepted policy/rating for 20A / 30A service.
- Initial battery charging may use AC-in-only mode with AC-out branch loads disconnected.

AC Safety/Protection Chain (What Must Exist)

- Upstream shore protection chain before MultiPlus AC-in: shore source/adapters -> portable EMS -> shore cord -> shore inlet -> 30A AC input breaker/disconnect.
- AC-out protection chain: MultiPlus AC-out-1 -> 10/3 feeder -> 30A AC-out main breaker -> 20A branch breakers -> GFCI receptacles.
- Single-enclosure DIN architecture with isolated AC-in and AC-out neutral paths and no neutral mixing.
- Continuous equipment grounding path from shore inlet through MultiPlus and branch circuits. The MultiPlus external M6 PE lug gets at least 4 mm², selected as 10 AWG green stranded copper, to a verified truck-chassis bond point; the aluminum shell gets a separate corrosion-compatible bond into the same equipment-ground network and is not the only PE path.
- Neutral/ground handling follows the MultiPlus internal relay: AC-out neutral bonds to chassis in inverter mode and opens that bond when shore is accepted. Do not add an always-bonded downstream neutral-ground bond in branch receptacle wiring.
- Keep PE and DC return roles separate: no MultiPlus case/PE jumper to Lynx or 12V negative. Confirm whether the Mechman is isolated- or case-grounded and prove no chassis path bypasses the SmartShunt before alternator commissioning.

AC Reference Basis (Manufacturer Guidance)

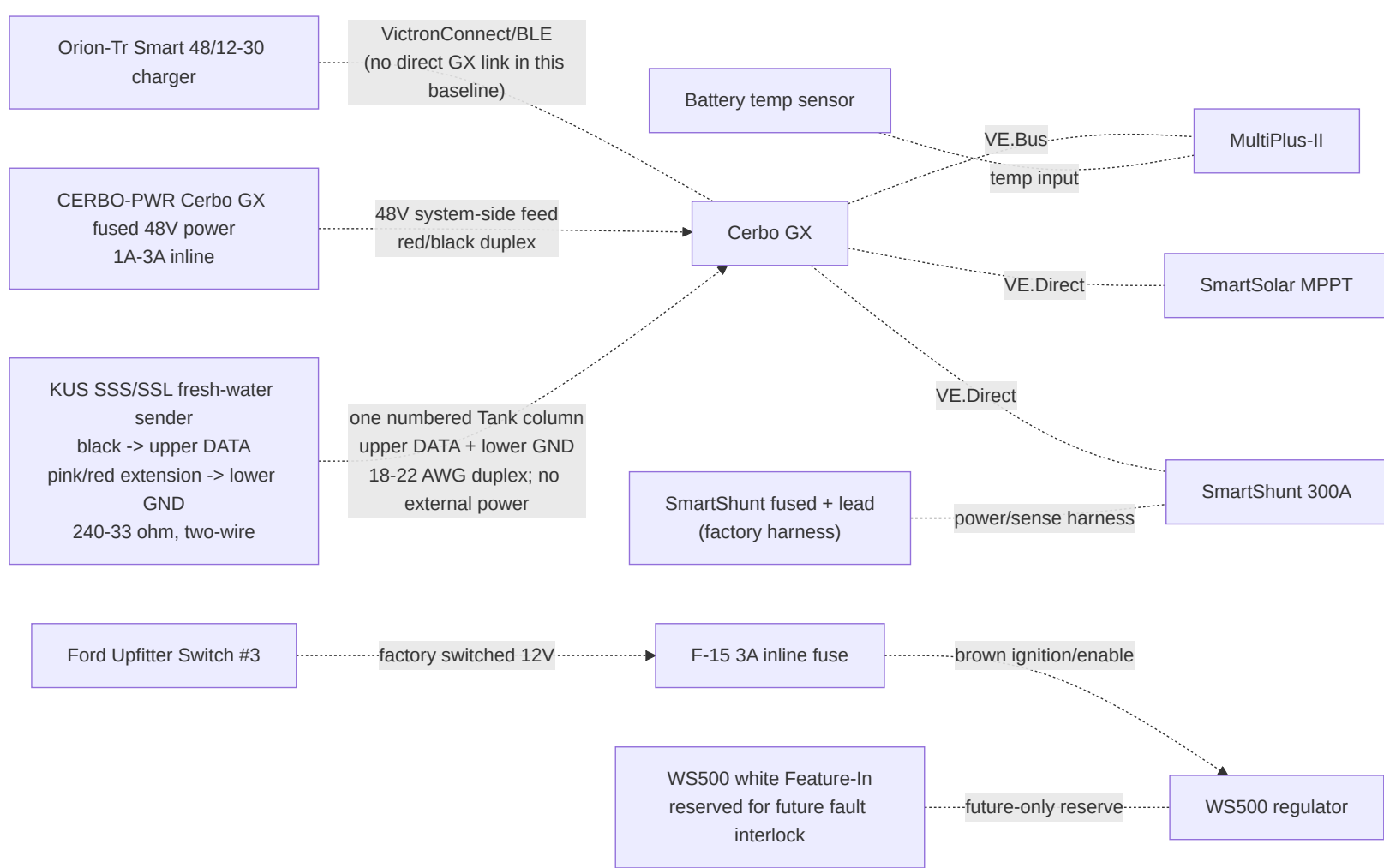
- Victron MultiPlus-II 120V installation guidance (AC-in breaker sizing, UL943-class residual-current protection on outputs, and AC-out-2 shore-only behavior): https://www.victronenergy.com/media/pg/MultiPlus-II_120V/en/installation.html
- Victron MultiPlus-II datasheet (48/3000/35-50 baseline model reference): <https://www.victronenergy.com/upload/documents/Datasheet-MultiPlus-II-inverter-charger-120V-EN.pdf>

AC/USB Baseline Locked For BOM

- Shore interface: 30A RV-style inlet baseline with adapter kit for 15A / 20A hookups.
- AC input protection: portable EMS + combined DIN enclosure + 30A AC input breaker/disconnect upstream of MultiPlus AC-in.
- AC-out-1 distribution: 30A AC-out main plus two protected 20A branches with GFCI-at-first-outlet strategy.
- Receptacle plan/as-built report (2026-08-27): Branch A office/driver and Branch B galley/passenger each have one first-in-chain 20A self-test GFCI plus one downstream duplex on the corresponding GFCI LOAD , for four duplex devices/eight plug positions total. Keep both 20A branches, 12 AWG , listed accessible boxes/covers/clamps, and whole-chain trip/reset/load testing.
- USB charging plan: 3 DC-fed USB PD station assemblies on separate 12V branches (1 office, 2 Galley), each standardized to 15A / 12 AWG . The third branch's actual fuse-panel slot/label is a field-audit item.

- AC-out-2 remains reserve-only in Phase 1 (labeled capped route; no energized branch hardware procured).

Monitoring and Control Topology



Fuse and Switch Housing Map (Where Each Item Is Physically Housed)

Item ID	Item value/type	Housing method	Location
F-01A	200A Class T (provisional)	Blue Sea Class T fuse block (110A-200A family)	Battery compartment near Battery A +
F-01B	200A Class T (provisional)	Blue Sea Class T fuse block (110A-200A family)	Battery compartment near Battery B +
F-01C	200A Class T (provisional)	Blue Sea Class T fuse block (110A-200A family)	Battery compartment near Battery C +
F-02	125A MEGA	Lynx integrated slot holder	Lynx Slot 1
F-03	60A MEGA (80V Victron replacement stock)	Lynx integrated slot holder	Lynx Slot 2
F-04	200A/80V MEGA	Lynx integrated slot holder	Lynx Slot 3 (dedicated alternator branch)
F-05	40A MEGA, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback	Lynx integrated slot holder	Lynx Slot 4; Orion 48v input feeder
F-06	Not installed / retired standalone input-fuse position	No holder	MIDI/FKS/DIN concepts superseded; do not stack after F-05

F-07	60A MEGA (80V Victron replacement stock)	Victron MEGA fuse holder	Electrical cabinet at Orion 12V + source end
F-09 PV string OCP position	0 planned for the purchased single-string Renogy 4S candidate; no peer string can backfeed it	Add listed gPV hardware only if received labels, conductor/connector limits, or applicable PV rules require it	Roof-entry/service area if reopened
F-10	Per branch (ATO/ATC)	Integrated blade sockets in generic 12V fuse block	Electrical cabinet
AUDIO-HU / 12V-12	15A source/head-unit branch; KMC2 harness also contains a 15A ATM fuse	12V fuse block branch plus KMC2 harness fuse	Electrical cabinet to driver-side DC shelf/source face
AUDIO-SUB	40A external fuse for Kicker PTRTP10 powered sub branch	Inline/MRBF/AFS/ANL-class holder matched to selected 4 AWG kit; use 40A, not a generic 100A kit fuse	Within about 18 in of the 12V source takeoff feeding the powered sub
F-11	100A class (12V buffer battery main)	Sealed inline MIDI/AMI/ANL holder	Within ~`7"`` of 12V buffer battery positive post
F-12/F-13-PHVAN	Littelfuse KLKD015.T, 15A/600VAC/DC fast-acting midget fuse	Eaton/Bussmann HEB-AA 600V water-resistant in-line holder	Immediately beside the Lynx, fed from the F-04 alternator/load-side stud through short 14 AWG pigtails; sealed splice to 16 AWG PH-VAN red; no panel, DIN rail, or enclosure
CERB0-PWR	1A-3A Cerbo GX power fuse	Small inline holder rated for the 48V bank maximum	Electrical cabinet near 48V system-positive takeoff; system side of disconnect preferred for bench shutdown
WS500 current-sense pair	No fuse; common battery-negative shunt with purple/high on battery-negative side and grey/low on SmartShunt/Lynx system side	Twist pair if extended; route away from noise; configure Shunt at Battery	Wakespeed shunt on the battery side of the hard-mounted SmartShunt
F-15	3A WS500 ignition/enable control fuse	Sealed inline ATC/ATO holder; 12V control circuit	Near Ford upfitter blunt-cut wire / WS500 control-wire handoff
SW-12V-BATT	Manual battery disconnect switch	Sealed rotary DC switch body	Electrical cabinet near 12V fuse-block main + stud for service access
OEM-SHUNT	External Victron-supplied inline fuse in red SmartShunt Vbatt+ cable	Inline holder in supplied red cable	Prefer battery-side positive if SOC continuity is desired with the main disconnect open; system side is acceptable for zero disconnect-off parasitic draw

Retired from active architecture:

- Obsolete pre-Mechman charger/fuse paths are removed from active board layout and labels.

Conductor Schedule (Start-to-Finish)

Segment ID	Circuit segment	Nominal voltage	Current basis	Overcurrent protection	Planned wire gauge	Estimate
C-01	Battery A + -> F-01A	48V	Battery branch, fuse-limited	F-01A 200A provisional	2/0 AWG	2.5 ft planned
C-02	Battery B + -> F-01B	48V	Battery branch, fuse-limited	F-01B 200A provisional	2/0 AWG	2.5 ft planned
C-02C	Battery C + -> F-01C	48V	Battery branch, fuse-limited	F-01C 200A provisional	2/0 AWG	2.5 ft planned

C-03	Class T outputs -> battery-side 48V + busbar -> disconnect input	48V	Combined trunk current	F-01A/B/C	2/0 AWG each branch	2.5 ft each branch
C-04	Disconnect output -> Lynx + bus	48V	Aggregate discharge design current (165A = F-02 125A + Orion F-05 40A)	Upstream Class T fuses	2/0 AWG	2.5 ft (ASSUMED)
C-05	Battery negatives -> battery-side 48V - busbar -> Wakespeed shunt battery/high side	48V	Mixed-path rollup: 3x battery-negative branches at 82.5A design each + NEGBUS_TO_WSSHUNT trunk at 165A aggregate	N/A (main negative path)	2/0 AWG each branch	2.5 ft each branch
C-06	Wakespeed shunt system/low side -> short 2/0 jumper -> SmartShunt battery side -> hard-mounted SmartShunt load side/Lynx - connection	48V	Aggregate return current	N/A	2/0 AWG jumper; no separate SmartShunt-to-Lynx cable	0.5 ft jumper
C-06A	Lynx positive tap -> SmartShunt positive sense/power lead	48V	Shunt electronics supply (very low current)	Factory inline fuse in OEM harness	OEM harness lead	2.5 ft (ASSUMED)
C-07	Lynx Slot 1 (F-02) -> MultiPlus DC+	48V	Inverter branch, fuse-limited	F-02 125A	2/0 AWG (manual minimum AWG 1 on short runs)	2.5 ft (ASSUMED)
C-08	MultiPlus DC- -> Lynx - bus	48V	Inverter return current	F-02 protects paired positive	2/0 AWG	2.5 ft (ASSUMED)
C-09	MPPT BAT+ -> Lynx Slot 2 (F-03)	48V	Controller output (45A max)	F-03 60A/80V Victron row 188	6 AWG	2.5 ft (ASSUMED)
C-10	MPPT BAT- -> Lynx - bus	48V	Controller return current	F-03 protects paired positive	6 AWG	2.5 ft (ASSUMED)
C-11	Secondary alternator B+ -> APM-48 -> Lynx Slot 3 (F-04)	48V	Published alternator output reaches about 145.7A	F-04 200A/80V	2/0 AWG	20 ft (ASSUMED)
C-12	Secondary alternator B- -> Lynx - bus (dedicated return)	48V	Alternator branch return current	F-04 paired	2/0 AWG	20 ft (ASSUMED)
C-13/C-14	Lynx Slot 4 (F-05) -> Orion 48V + input	48V	Orion feeder, fuse-limited	F-05 40A MEGA, >=58VDC	Existing 6 AWG direct; 10 AWG adequate if ever replaced	2.5 ft (ASSUMED)
C-15	Orion 48V - input -> Lynx - bus	48V	Orion input return current	Orion input positive protection paired	6 AWG	2.5 ft (ASSUMED)
C-18	Orion 12V + -> F-07 -> 12V fuse block	12V	Charger output path (30A continuous, 60A fuse)	F-07 60A	6 AWG planned (8 AWG minimum per Orion)	2.5 ft (ASSUMED)

	main + stud				table)	
C-19	Orion 12V - -> 12V fuse block integrated - bus / main - stud	12V	Charger output return	F-07 protects paired positive	6 AWG	2.5 ft (ASSUMED)
C-19A	12V buffer battery + -> F-11 -> SW-12V-BATT -> 12V fuse block main + stud	12V	Buffer source path and service isolation path	F-11 100A class	4 AWG planned	2.5 ft (ASSUMED)
C-19B	12V buffer battery - -> 12V fuse block integrated - bus / main - stud	12V	Buffer battery return path	N/A (paired with C-19A)	4 AWG planned	2.5 ft (ASSUMED)
C-20	12V panel -> purchased Starlink DC converter	12V	Purchased converter is the planned primary path; supplied AC power remains fallback until bench/load acceptance	Dedicated 20A ATO/ATC branch fuse planned; size remains limited by received input lead and installed conductor	Red/yellow marine duplex planned; 12 AWG only within about 6 ft one-way, otherwise 10 AWG	HOLD energy measured re-heat, and re-
C-21	12V panel -> Fridge	12V	Branch load	F-10 15A	12 AWG duplex	12 ft (ASSUMED prior voltage)
C-22	12V panel -> LF Bros diesel-heater harness	12V	Startup/cooldown branch; keep energized through controller-commanded shutdown	F-10 15A (retain any supplied harness fuse; verify coordination)	12 AWG duplex	8 ft (ASSUMED)
C-23	12V panel -> Water pump	12V	Branch load	F-10 10A	14 AWG duplex	8 ft (ASSUMED)
C-24	12V panel -> CO + propane detector	12V	Branch load	F-10 3A	18/2	8 ft (ASSUMED)
C-25	12V panel -> LED lights + dimmer (Hiatus pre-installed)	12V	Branch load	F-10 5A	18/2	8 ft (ASSUMED)
C-26	48V system positive/negative -> CERBO-PWR -> Cerbo GX power input	48V	Cerbo electronics feed (~3W)	CERBO-PWR 1A-3A inline fuse	18 AWG red/black duplex acceptable	2.5 ft (ASSUMED)
C-27	Purchased Renogy 4S string -> two-polarity roof transition box -> continuous front-gap retractile -> fixed-shell premium gland -> two accessible sealed splices -> stationary 12/2 -> two-pole load-	78.0V Vmp; 95.6V Voc; 114.86V nominal-coefficient cold Voc at -40C	8.98A Imp; 9.50A Isc STC; 14.8A conservative conductor screen	No F-09 presently planned for one string; reopen only from received-label/code review	Preferred 12 AWG x 2 fixed/moving PV path after final route/derating proof; moving cord requires written >=150VDC, coiled ampacity, outdoor/wet/UV/cold/flex proof. Current envelope is ~24 in down / ~52 in up; research 60-66 in rated overall. No four-to-one branch/combiner.	No exterior lengths/OD/ remain on fu restart/track

	break -> MPPT PV input					
C-28	Shore source/adapters -> portable EMS -> shore cord -> shore inlet -> combined AC DIN enclosure / AC input breaker	120VAC	Source-limited shore current (adapter-constrained at source)	30A AC input breaker/disconnect baseline with source-current-limit settings policy	10/3 shore feed to inlet/AC-in area	8 ft (ASSUMED)
C-29	AC input breaker/disconnect -> MultiPlus AC-in	120VAC	MultiPlus AC input current (30A hardware basis)	Upstream 30A AC breaker/disconnect (C-28)	10 AWG stranded AC conductors	2.5 ft (ASSUMED)
C-30	MultiPlus AC-out-1 -> combined AC DIN enclosure / AC-out main breaker	120VAC	Inverter-backed AC-out feeder current (30A system cap)	30A AC-out main breaker	10/3 stranded AC feeder	2.5 ft (ASSUMED)
C-31	Branch A -> office/driver GFCI receptacle	120VAC	Branch load (office monitor/chargers/general outlet use)	20A branch breaker + GFCI receptacle	12/3 stranded AC branch cable	15 ft (ASSUMED)
C-32	Branch B -> galley/passenger GFCI receptacle	120VAC	Branch load (galley/general high-draw outlet; induction/Ninja SP151 sequenced)	20A branch breaker + GFCI receptacle	12/3 stranded AC branch cable	15 ft (ASSUMED)
C-33	MultiPlus AC-out-2 (reserve-only) -> capped route for future shore-only branch	120VAC	N/A in Phase 1 (route reserved only)	N/A in Phase 1 (no energized branch hardware)	12 AWG stranded AC conductors (reserve path only)	15 ft (ASSUMED)
C-34	12V panel -> USB PD station branch (office zone)	12V	One Acegoo 118W station	F-10 branch fuse (15A)	12 AWG duplex baseline	5 ft (ASSUMED)
C-35	12V panel -> USB PD station branch (galley zone)	12V	Galley charging branch (65W class USB-C plus USB-A/C loads)	F-10 branch fuse (15A)	12 AWG duplex baseline	8 ft (ASSUMED) rough-in for
C-36	12V panel -> Maxxair fan (Hiatus pre-installed)	12V	Roof ventilation branch	F-10 branch fuse (10A)	14 AWG duplex baseline	8 ft (ASSUMED)
C-37	12V panel -> DC ambient/cabinet LED strips (planned Govee)	12V	Branch load	F-10 branch fuse (5A)	18/2 baseline	8 ft (ASSUMED)
C-38	F-04 alternator/load-side stud -> Eaton/Bussmann HEB-AA + Littelfuse KLKD015.T ->	48V bank	Combined regulator-power / positive-voltage-sense feed	F-12/F-13-PHVAN (15A)	Short 14 AWG pigtails, sealed transition to 16 AWG harness	In-line beside

	WS500 PH-VAN red lead					
C-39	Retired separate WS500 positive-sense conductor	N/A	Not installed with confirmed PH-VAN harness	None	N/A	N/A
C-40	WS500 current-sense pair to common battery-negative shunt: purple/high battery-negative side, grey/low SmartShunt/Lynx system side	low-current sense	Net battery-current feedback	No fuse; twist pair if extended; configure Shunt at Battery	Harness lead	Local near h
C-41	Ford Upfitter #3 output -> F-15 -> WS500 brown ignition/enable wire	12V control lead	Manual alternator-enable signal only	F-15 (3A)	16 AWG TXL/GXL	6 ft (ASSUMED
C-42	12V panel -> Kicker 46KMC2 media center/source unit	12V	Source/head-unit branch (15A max; KMC2 manual shows 15A ATM)	F-10/AUDIO-HU 15A branch plus KMC2 harness fuse	12 AWG duplex if kept near 5 ft; use 10 AWG if longer	5 ft (ASSUMED
C-43	12V source/main + stud -> AUDIO-SUB source fuse -> Kicker 49PTRTP10 powered sub +	12V	Powered sub branch; PTRTP10 manual external fuse value	AUDIO-SUB 40A	4 AWG tinned/OFC	8 ft (ASSUMED junction)
C-44	Kicker 49PTRTP10 powered sub - -> 12V negative bus/main - stud	12V	Powered sub return	AUDIO-SUB protects paired positive	4 AWG tinned/OFC	8 ft (ASSUMED
C-45	KMC2 remote turn-on output -> PTRTP10 remote input	12V low-current control	Remote amp/sub enable	KMC2/source branch protected	18 AWG	8 ft (ASSUMED
C-46	KMC2 RCA line-out -> PTRTP10 RCA input	low-level audio signal	Subwoofer signal	N/A	shielded 2-channel RCA	measured ro
C-47L/R	KMC2 front speaker outputs - > left/right Kicker CSC67 speakers	speaker-level audio	One 4 ohm speaker per KMC2 channel	KMC2 internal protection / AUDIO-HU source branch	16 AWG marine speaker wire	10-12 ft eac
C-48	KUS SSS/SSL sender black signal + pink return -> matching signal/negative pair in Cerbo GX MK2 Tank 1 column	Cerbo low-current resistive-sender excitation	Fresh-water level, US 240-30 ohm	No external fuse or power feed; do not share chassis return	18-22 AWG tinned duplex, ferrules at Cerbo, sealed pigtail splices at sender	Measure aft (ASSUMED <10

C-49	12V panel -> second Galley USB PD station	12V	Third independent PD branch	F-10 branch fuse (15A); exact panel slot/label to verify in field	12 AWG duplex	Owner report final length
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Wiring Validation Worksheet (Estimate Pass, 2026-02-18)

Calculation basis for drop screening:

- $V_{drop} = I * (2 * L_{one_way} * R_{per_ft})$
- Resistance basis used in this pass (ohm/ft): 2/0=0.0000779 , 6 AWG=0.0003951 , 4 AWG=0.0002485 , 12 AWG=0.001588 , 14 AWG=0.002525 , 18/2=0.006385 , 10 AWG=0.000999 .
- Design targets: <=2% on major 48V trunks, <=3% on planned 12V /AC branches.
- C-05 is a rollup row that includes both branch and trunk return paths; voltage-drop screen shown is the conservative interim worst-case (155A) within that rollup.

Circuit ID	From	To	Fuse	Current basis	Gauge	Estimated one-way length	Voltage drop %
C-01	Battery A +	F-01A	F-01A 200A	77.5A interim design branch share	2/0 AWG	2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced	0.059% @ 51.2V planning placeholder
C-02	Battery B +	F-01B	F-01B 200A	77.5A interim design branch share	2/0 AWG	2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced	0.059% @ 51.2V planning placeholder
C-02C	Battery C +	F-01C	F-01C 200A	77.5A interim design branch share	2/0 AWG	2.5 ft planning placeholder; field layout may use unequal positive lengths if total loop path is balanced	0.059% @ 51.2V planning placeholder
C-03	Class T load studs	48V + bus / disconnect input	F-01A/B/C	77.5A interim per branch	2/0 AWG	2.5 ft each (x4 conductors)	0.059% @ 51.2V

C-04	Disconnect output	Lynx + bus	Upstream Class T	155A interim / 145A final aggregate	2/0 AWG	2.5 ft	0.12% @ 51.2V interim
C-05	Battery - branches	Wakespeed shunt battery/high side via 48V - bus	N/A	77.5A per battery-negative branch; row rollup also includes one 155A interim trunk (NEGBUS_TO_WSSHUNT)	2/0 AWG	2.5 ft each (x4 conductors)	0.12% @ 51.2V (worst-case rollup)
C-06	Wakespeed shunt system/low side	SmartShunt battery side; SmartShunt load side remains hard-mounted to Lynx -	N/A	155A interim / 145A final aggregate return	Short 2/0 AWG jumper	0.5 ft placeholder	0.024% @ 51.2V interim
C-06A	Lynx positive tap	SmartShunt sense/power lead	OEM inline fuse	OEM harness current	OEM harness	2.5 ft	N/A (low-current OEM lead)
C-07	Lynx Slot 1 DC+	MultiPlus DC+	F-02 125A	125A	2/0 AWG	2.5 ft	0.10% @ 51.2V
C-08	MultiPlus DC-	Lynx - bus	F-02 paired	125A	2/0 AWG	2.5 ft	0.10% @ 51.2V
C-09	MPPT BAT+	Lynx Slot 2	F-03 60A (80V Victron row 188)	45A	6 AWG	2.5 ft	0.17% @ 51.2V
C-10	MPPT BAT-	Lynx - bus	F-03 paired	45A	6 AWG	2.5 ft	0.17% @ 51.2V
C-11	Secondary alternator B+	Lynx Slot 3 via APM-48	F-04 200A/80V	145.7A published-curve screen	2/0 AWG	20 ft	<0.80% @ 58.4V
C-12	Secondary alternator B-	Lynx - bus (dedicated return)	F-04 paired	145.7A published-curve screen	2/0 AWG	20 ft	<0.80% @ 58.4V
C-13/C-14	Lynx Slot 4	Orion 48V +	F-05 40A MEGA, >=58VDC	40A fuse basis	Existing 6 AWG direct	2.5 ft	0.16% @ 51.2V
C-15	Orion 48V -	Lynx - bus	Orion input positive protection paired	40A fuse basis	6 AWG	2.5 ft	0.16% @ 51.2V
C-18	Orion 12V +	Fuse block main + stud	F-07 60A/80V Victron row 188	30A	6 AWG	2.5 ft	0.49% @ 12V
C-19	Orion 12V -	Fuse block integrated - bus / main - stud	F-07 paired	30A	6 AWG	2.5 ft	0.49% @ 12V
C-19A	Buffer battery +	Fuse block main + stud (via F-11/SW)	F-11 100A	50A design	4 AWG	2.5 ft	0.52% @ 12V
C-19B	Buffer battery -	Fuse block integrated - bus / main - stud	N/A	50A design	4 AWG	2.5 ft	0.52% @ 12V
C-20	12V fuse panel	Purchased Starlink DC converter	Dedicated 20A ATO/ATC	Measure received input/current	12 AWG marine duplex only	Measure	Recalculate from

			branch planned; not installed	behavior	within about 6 ft one-way; otherwise 10 AWG		measured route/current
C-21	12V fuse panel	Fridge	F-10 15A	7A	12 AWG duplex	12 ft	2.23% @ 12V
C-22	12V fuse panel	LF Bros diesel- heater harness	F-10 15A + supplied harness fuse if present	10A startup screen	12 AWG duplex	8 ft assumed	2.12% @ 12V
C-23	12V fuse panel	Water pump	F-10 10A	7A	14 AWG duplex	8 ft	2.36% @ 12V
C-24	12V fuse panel	CO+propane detector	F-10 3A	0.2A	18/2	8 ft	0.17% @ 12V
C-25	12V fuse panel	LED lights + dimmer (Hiatus pre-installed)	F-10 5A	5A	18/2	8 ft	4.26% @ 12V
C-26	48V system feed	Cerbo GX	CERB0-PWR 1A- 3A	~3W	18 AWG duplex acceptable	2.5 ft	N/A (low- current electronics feed)
C-27	Purchased Renogy 4S string + two- polarity transition box + continuous front-gap retractile + premium gland + stationary 12/2 + two-pole disconnect	MPPT PV input	No F-09 presently planned	9.50A I _{sc} STC / 14.8A conservative conductor screen	Preferred 12 AWG x 2 after route/derating proof	Full route pending; 20 ft one-way screen	0.94% at 9.5A / 78V for 12 AWG
C-28	Shore inlet path	AC input breaker/disconnect	Source- limited AC OCP	30A hardware basis	10/3	8 ft	0.40% @ 120VAC
C-29	AC input breaker/disconnect	MultiPlus AC-in	Upstream AC OCP	30A hardware basis	10 AWG AC	2.5 ft	0.12% @ 120VAC
C-30	MultiPlus AC-out-1	AC-out main breaker	30A AC-out main	30A hardware basis	10/3	2.5 ft	0.12% @ 120VAC
C-31	Branch A	Office/driver receptacle chain	20A branch OCP	20A	12 AWG AC	15 ft	0.79% @ 120VAC

C-32	Branch B	Galley/passenger receptacle chain	20A branch OCP	20A	12 AWG AC	15 ft	0.79% @ 120VAC
C-33	MultiPlus AC-out-2	Reserve-only capped route	N/A in Phase 1	N/A	12 AWG AC (reserve path only)	15 ft	0.60% @ 120VAC (future-use screen)
C-34	12V fuse panel	Office USB PD station	F-10 15A	15A design cap	12 AWG duplex	5 ft	1.99% @ 12V
C-35	12V fuse panel	Galley USB PD station	F-10 15A	8A expected	12 AWG duplex	8 ft	1.69% @ 12V
C-49	12V fuse panel	Second Galley USB PD station	F-10 15A; slot to verify	15A design cap	12 AWG duplex	Field routed; measure	Recalculate from measured length
C-36	12V fuse panel	Maxxair fan (Hiatus pre-installed)	F-10 10A	4A expected	14 AWG duplex	8 ft	1.35% @ 12V
C-37	12V fuse panel	DC ambient/cabinet LED strips (planned Govee)	F-10 5A	5A design cap	18/2	8 ft	4.26% @ 12V
C-38	F-04 alternator/load-side stud	WS500 PH-VAN short red lead	F-12/F-13-PHVAN Littelfuse KLKD015.T 15A/600VAC/DC	combined regulator-power / voltage-sense feed at 48V bank voltage	14 AWG pigtails to 16 AWG harness	Local	N/A (harness-limited)
C-39	Retired separate positive-sense source	Not installed	None	superseded by confirmed PH-VAN combined lead	N/A	N/A	N/A
C-40	Common battery-negative Wakespeed shunt: purple/high battery-negative side, grey/low SmartShunt/Lynx system side	WS500 current-sense input	No fuse	low-current sense; twist pair if extended	Harness lead	Local	N/A (harness-limited)
C-41	Ford Upfitter #3 output	WS500 brown ignition/enable input via F-15	F-15 3A	manual low-current control only	16 AWG TXL/GXL	6 ft	N/A (control circuit)
C-42	12V panel	Kicker 46KMC2 media center	AUDIO0-HU 15A branch + KMC2 15A ATM harness fuse	15A max fuse basis	12 AWG duplex	5 ft	1.99% @ 12V
C-43/C-44	12V source/main studs	Kicker 49PTRTP10 powered sub	AUDIO0-SUB 40A external source fuse	40A manual fuse basis	4 AWG positive + matching return	8 ft	1.33% @ 12V
C-45	KMC2 remote output	PTRTP10 remote input	KMC2/source branch	low-current remote	18 AWG	8 ft	N/A

			protected				
C-46	KMC2 RCA line-out	PTRTP10 RCA input	N/A	low-level signal	shielded RCA	measured route	N/A
C-47L/R	KMC2 front speaker outputs	Left/right CSC67 speakers	KMC2/source branch protected	one 4 ohm speaker per channel	16 AWG marine speaker wire	10-12 ft each	N/A

Wire Rollup (No-Padding Purchase Baseline)

Gauge / cable family	Estimated total	Source circuits	BOM row
2/0 AWG red	42.5 ft	C-01, C-02, C-02C, C-03, C-04, C-07, C-11	28
2/0 AWG black	33.0 ft	C-05, C-06, C-08, C-12	28
6 AWG red	7.5 ft	C-09, C-13/C-14, C-18	29
6 AWG black	7.5 ft	C-10, C-15, C-19	29
4 AWG red	10.5 ft (2.5 ft existing + 8 ft audio sub planning branch)	C-19A, C-43	30, 192
4 AWG black	10.5 ft (2.5 ft existing + 8 ft audio sub return)	C-19B, C-44	30, 192
12 AWG PV cable allowance	Route length pending	C-27 purchased Renogy 4S; three series mates, two-polarity roof transition, continuous front-gap wet/flex retractile, stationary straight lower gland tail, then two sealed splices to stationary 12/2; no four-to-one combiner or F-09 presently planned	31, 106, 121
14 AWG duplex	16 ft	C-23, C-36	32
18/2	24.0 ft plus separate Cerbo low-current duplex	C-24, C-25, C-37; Cerbo C-26 uses 48V fused low-current duplex	33 / low-current stock
12 AWG AC branch cable	30 ft (C-33 excluded in Phase 1)	C-31, C-32	113
10/3 shore + AC-in/out feed	13 ft	C-28, C-29, C-30	114
12 AWG fridge/heater/USB/audio branch mix	Fridge: 12 ft; heater: 8 ft; USB: 5 ft office + 8 ft Galley A + measured Galley B field run; audio source: 5 ft short-run assumption	C-21, C-22, C-34, C-35, C-49, C-42	116, 193 / field stock
Camper audio signal/speaker/control wiring	RCA measured route, 18 AWG remote, and 16 AWG marine speaker wire runs	C-45, C-46, C-47L/R	193
WS500 harness/sense leads	Included in selected kit/harness set; C-39 is retired under PH-VAN	C-38, C-40	168, 171
16 AWG TXL/GXL control wire	6 ft	C-41	176

Notes:

- This table is a base estimate only; it intentionally excludes order padding and termination waste.
- Apply personal order overage at checkout based on actual spool cut increments and routing confidence.
- Parallel bank balancing is locked by similar total loop resistance per battery. Positive-only leads may differ if the paired negative path offsets the difference; verify with final measured lengths and, if needed, clamp-current checks.

3x Battery Bank Bench-Build Cut List (2/0 AWG)

Purpose: make the bench build orderable without needing final camper run lengths. Treat lengths below as *bench module* lengths only; final install harnesses should be re-cut after layout freeze.

Assumptions:

- Battery terminals are M8 (verify your battery stud size before ordering lugs).
- Class T fuse blocks and battery-side busbars use 3/8" studs (treat as M10 lugs unless your specific hardware differs).
- Lynx Distributor is the M10 model (main connections M10 ; internal/fuse studs may still be M8 depending on the position).

Cable ID	Qty	From -> To	Color	Gauge	Est. one-way length	Lug A	Lug B
BATT+_A/B/C	3	Battery + -> Class T block line side	red	2/0	2.5 ft each (ASSUMED)	M8	M10
FUSE_TO_POSBUS_A/B/C	3	Class T block load side -> 48V + busbar	red	2/0	2.5 ft each (ASSUMED)	M10	M10
POSBUS_TO_DISC	1	48V + busbar -> disconnect input	red	2/0	2.5 ft (ASSUMED)	M10	M10
DISC_TO_LYNX+	1	disconnect output -> Lynx + input	red	2/0	2.5 ft (ASSUMED)	M10	M10
BATT-_A/B/C	3	Battery - -> 48V - busbar	black	2/0	2.5 ft each (ASSUMED)	M8	M10
NEGBUS_TO_WSSHUNT	1	48V - busbar -> Wakespeed shunt battery/high side	black	2/0	2.5 ft (ASSUMED)	M10	Verify shunt stud
WSSHUNT_TO_SMARTSHUNT	1	Wakespeed shunt system/low side -> SmartShunt battery side	black	2/0	0.5 ft placeholder	Verify shunt stud	M10
SMARTSHUNT_TO_LYNX-	0 cable	SmartShunt load side remains hard-mounted directly to Lynx - input	—	—	Existing stud connection	M10	M10
LYNX_SLOT1_TO_MULTI+	1	Lynx Slot 1 DC+ -> MultiPlus DC+	red	2/0	2.5 ft (ASSUMED)	M8	M8
LYNX_TO_MULTI-	1	Lynx - -> MultiPlus DC-	black	2/0	2.5 ft (ASSUMED)	M8	M8

Locked balancing rule for the 3x parallel bank:

- Keep each battery path's total positive + negative loop resistance similar.
- Equal positive-only leads are not required when the negative lead lengths intentionally offset the positive lead differences.
- Keep the same cable family, lug geometry, fuse/holder family, and termination quality across all three battery paths.
- Verify sharing with clamp-current checks under charge/load if final measured path lengths differ materially.

Torque reference (verify against your exact manuals/hardware):

- MultiPlus-II DC terminals: 12 Nm (M8 nut) per Victron installation guidance.
- SmartShunt shunt bolts: verify torque for the installed 300A model per Victron installation guidance.
- Lynx Distributor M10 model: M10 nuts 33 Nm (older serials may be lower), and M8 nuts 14 Nm per Victron Lynx installation guidance.

Additional Components Included In Topology Scope

- 48V disconnect (275A)
- Pre-charge resistor (commissioning/soft-charge aid before connecting large DC loads)
- Battery-side 48V + combine busbar (after Class T fuses)
- Battery-side 48V - combine busbar (battery-only, before SmartShunt)
- 12V fuse block main + stud used as source-combine point (Orion + buffer battery feed)
- 12V fuse block integrated negative bus/main - used as shared return point
- 12V buffer battery main fuse (F-11) and manual disconnect switch (SW-12V-BATT)
- Shore AC inlet + cord/adapter interface hardware

- Portable EMS in source-side shore path before camper inlet
- Single combined 6-way AC DIN enclosure with isolated AC-in and AC-out neutral paths plus common equipment grounding/PE handling
- AC-out 30A main breaker plus 20A / 20A branch breaker and GFCI receptacle hardware
- Receptacle boxes + 120V outlets (current purchased baseline 2 first-in-chain GFCI receptacles/covers in row 15 ; active row 112 now records six purchased old-work boxes for field-fit downstream devices, but device count/covers/clamps/fill/access/testing remain open)
- AC-out-2 reserve-only capped route (no energized Phase 1 branch hardware)
- USB PD station branch hardware (3 stations / independent routes owner-reported 2026-08-05 : 1x office and 2x Galley; row 115 owns the exact purchased Acegoo evidence while the additional station purchase provenance remains untracked)
- Camper audio hardware: Kicker 46KMC2 source branch, Kicker 49PTRTP10 powered sub branch, AUDIO-HU 15A source protection, AUDIO-SUB 40A source fuse, RCA/speaker/remote wiring
- Battery temperature sensor wiring to inverter/monitoring path
- SmartShunt fused positive sense/power lead (factory harness)
- Cerbo GX CERB0-PWR small inline fused 48V power feed
- Ford Upfitter #3 control lead and local F-15 inline fuse for the WS500 brown ignition/enable wire

Assumptions (Explicit)

1. Cable sizing assumes fine-strand copper conductors (OFC welding-cable baseline for high-current DC paths), enclosed vehicle routing, and the estimated one-way lengths listed in this document.
2. Voltage-drop design intent used here: $\leq 2\%$ on major 48V power runs and $\leq 3\%$ on 12V branch circuits.
3. The purchased Renogy panels are currently screened as one 4S string and presently use no F-09 series fuse because no peer string can backfeed them. Received labels must confirm Isc and maximum series-fuse rating; reopen OCP if conductor/connector limits or applicable PV rules require it.
4. Cerbo GX feed is now a small inline fused 48V feed (CERB0-PWR) from the system/load side of the main disconnect during bench commissioning, so the Cerbo powers down with the house system.
5. Orion branch uses one source-side fuse only: F-05 40A MEGA, body-marked at least 58VDC , in Lynx Slot 4 feeding the existing 6 AWG input pair directly. The 58VDC minimum is tied to the locked 56.8V charge ceiling; use Victron CIP138040020 40A/80V if replacement is needed. Standalone F-06 is retired; do not add an inline fuse after the Lynx slot.
6. No low-voltage-disconnect (LVD) automation is included in Phase 1; protection is source fusing plus manual SW-12V-BATT isolation.
7. Alternator architecture lock is dedicated 48V secondary alternator path (Mechman + WS500 + APM-48) with F-04 200A/80V ; obsolete pre-Mechman engine-bay fuse paths are removed from active layout.
8. F-01A/B/C are provisionally set to 200A pending final 51.2V battery datasheet/manual confirmation; if validated limits are lower, shift to 175A .
9. 2/0 cable quantity planning baseline in this pass is 77.5 ft total no-padding (42.5 ft red + 35.0 ft black); user-applied order padding is intentionally deferred to checkout.
10. Manual alternator shutdown baseline is Ford Upfitter #3 feeding the WS500 brown ignition/enable wire through F-15 ; WS500 white Feature-In remains a future-only reserve for automatic interlock work.
11. Camper audio is a 12V accessory load. KMC2 source branch is 15A ; PTRTP10 sub branch is 40A with 4 AWG power and return. Audio peaks can exceed Orion continuous output and rely on the 12V buffer battery, so first loud testing should watch 12V voltage/SOC.

Completion Status

- DC/PV topology is complete for current BOM scope and load model scope.
- AC hierarchy is complete at architecture level, including transfer behavior, branch strategy, and protection chain.
- Full-circuit estimate pass is now documented with run lengths, voltage-drop screening, and purchase rollups.
- Camper audio branch added as a DC-first 12V accessory package: Kicker 46KMC2 source branch plus Kicker 49PTRTP10 powered sub branch; detailed product/routing notes live in docs/implementation/CAMPER_audio_system.md .
- Remaining work is final measured-length replacement and SKU-level closeout before final cut-to-length harness production.